TECHNICAL DOSSIER VOLUME 1

SURGCLEAR PLATFORM OPERATIONAL & TECHNICAL ARCHITECTURE

The Multi-Stakeholder Transaction Engine: Unifying Specialist Practices, Switching Houses, Medical Schemes, Gap Underwriters, and Hospital Admissions.

HL7 FHIR R4 Specification, Zero-Float ZAR Cent Math, POPIA Compliance & Digital Check-In Protocols

DOCUMENT TYPE
System Architecture & Technical Specifications

INTEROPERABILITY STANDARD
HL7 FHIR R4 (Institutional Claim, Patient, Coverage)

INTEGRATION PROTOCOLS
MediSwitch EDI, Healthbridge Switch, SpesNet, REST API

COMPLIANCE STANDARD
POPIA (Act 4 of 2013), ECTA Section 13 Digital Consent

SECURITY MODEL
AES-256 field encryption (pgcrypto), TLS 1.3, Signed QR Tokens

VERSION BASELINE
v2.4 Production Standard (2026 Statutory Rules)

1. The Multi-Stakeholder Ecosystem Model

Private surgery in South Africa involves five disconnected participants: the **Specialist Surgeon/Anaesthetist**, the **Medical Scheme**, the **Gap Cover Underwriter**, the **Private Hospital Facility**, and the **Patient**. Because no unified data interchange exists, each participant operates in a data silo.

> **Architectural Mission:** SurgClear acts as the real-time financial clearance orchestrator. Rather than waiting for post-operative paper claims and billing disputes, SurgClear aggregates clinical coding, scheme rules, gap coverage, and patient consent *before the patient ever sets foot in theatre*.

< 30s	100%	< 45s
SPECIALIST BOOKING	ZERO-FLOAT MATH	HOSPITAL QR CHECK-IN

End-to-End Stakeholder Data Pipeline

Phase	Stakeholder Action	Technical Data Flow	Platform Output
1. Initiation	Surgeon / Receptionist enters SA ID & ICD-10/RPL codes in Practice Cockpit.	Luhn verification, EDI switch query via MediSwitch / Healthbridge.	Matched Scheme Plan, MSA balance, Gap policy, and base tariffs.
2. Clearance	Cost Engine calculates shortfalls, DSP penalties, and Gap offsets.	Integer ZAR cent algorithm evaluates 100%-300% RPL rates against 2026 R223k OAL.	Complete financial liability split (Scheme vs Gap vs Patient).
3. Mobile Pass	Patient receives secure magic WhatsApp/SMS link on smartphone.	Mobile PWA loads encrypted episode, displays meal options and digital consent.	Signed ECTA consent, meal choice (Halal/Kosher/Soft), upfront co-pay receipt.
4. Pre-Auth	Authoritative pre-authorisation request dispatched to medical scheme.	HL7 FHIR R4 Institutional Claim resource serialized with BHF codes.	Authoritative Scheme Pre-Auth # (`PA-xxxxxx-ZA`) + Bed Approval.
5. Check-In	Hospital reception scans patient QR boarding pass (`ADM-xxxx-JHB`).	Cryptographic HMAC token validated against Hospital EMR / HIS interface.	45-second admission; kitchen notified of meal; co-pay marked cleared.
6. Auto-Gap	Post-op electronic Remittance Advice (RA) received from medical scheme.	SurgClear packages hospital invoice, doctor bills, and RA into Gap claim.	Auto-adjudicated Gap Claim filed directly to Sirago/Turnberry with zero paperwork.

2. Core Technical Architecture & Data Security

A. Zero-Floating-Point Currency Engine

Financial rounding errors are unacceptable in healthcare billing. SurgClear enforces a strict **integer ZAR cents standard** across all schemas, APIs, and calculation routines. Every amount is stored as an integer (e.g., R12,650.00 = `1265000`). Floating-point arithmetic is strictly prohibited in the engine, eliminating IEEE-754 precision drift.

B. HL7 FHIR R4 Institutional Claim Implementation

SurgClear generates authoritative HL7 FHIR R4 payloads conforming to South African Health Norms (URN `urn:oid:2.16.710.1`). The pre-authorisation payload maps directly to an institutional `Claim` resource:

- **Identifier:** Episode reference UUID with practice BHF registration.
- **Provider:** Specialist doctor registered with the Board of Healthcare Funders (BHF PCNS).
- **Facility:** Licensed hospital facility code (Netcare, Life Healthcare, Mediclinic, Busamed, Cure Day).
- **Diagnosis Array:** Validated WHO ICD-10 clinical diagnostic codes (e.g. `M23.2` Meniscus Derangement).
- **Procedure Item Array:** South African Reference Price List (RPL) surgical tariff codes (e.g. `2443` Knee Arthroscopy) with approved monetary values.

C. POPIA Compliance & Field-Level Cryptography

Under the Protection of Personal Information Act (POPIA Act 4 of 2013), clinical diagnoses, medical aid details, and national identification numbers constitute *Special Personal Information*. SurgClear implements a Defense-in-Depth security framework:

- **Field-Level Encryption (PostgreSQL pgcrypto):** SA ID numbers and member numbers are encrypted at rest using AES-256 with isolated key rings.
- **Row-Level Security (RLS):** Multi-tenant isolation guarantees that specialist practices and hospital facilities only access episodes explicitly associated with their BHF numbers.
- **Advanced Electronic Signatures (ECTA Section 13):** Patient surgical indemnities, POPIA disclosures, and financial acknowledgments are cryptographically signed with timestamped IP, user-agent, and canvas hash logs.
- **High-Contrast Offline QR Tokens:** The patient admission boarding pass uses an HMAC-SHA256 signed compact token (`ADM-xxxx-JHB`) that can be scanned even if the patient's phone has intermittent cellular reception.

> **Infrastructure Resilience:** SurgClear's serverless Next.js edge deployment on Vercel paired with Supabase PostgreSQL delivers sub-15ms regional latency across South Africa (via African edge nodes in Johannesburg and Cape Town), with 99.99% uptime SLA.